PROJECT WORKFLOW

Intelligent Traffic Management and Emergency Routing System

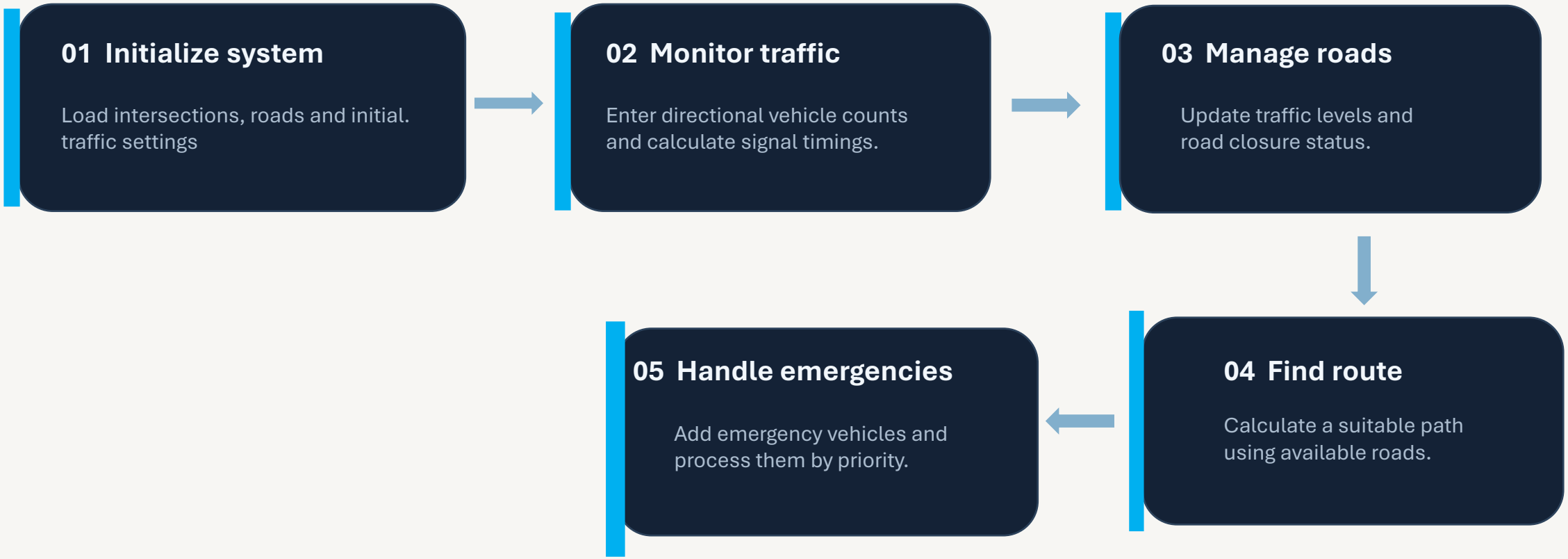
Workflow of traffic monitoring, road management, route calculation and emergency vehicle handling.

OOP PROJECT PRESENTATION

Santoshi swain

Overall Project Workflow

The system moves from initialization to processing and displaying results.



1. System Initialization & Traffic Monitoring

Prepare the network, then adapt signal timing using vehicle counts.

Initialize road network

- Create intersections
- Store connected roads and distances
- Set initial traffic levels
- Mark roads open or closed

Enter vehicle counts

- North direction
- South direction
- East direction
- West direction

Calculate signal timing

- Process directional counts
- Calculate green time
- Display updated timings

Output: updated signal timings for the four directions.

2. RoadNetwork Management

Update road conditions before calculating a route.

Select a road

Choose the starting and ending intersections of a connected road.

Update traffic

Set the traffic level for the selected road.

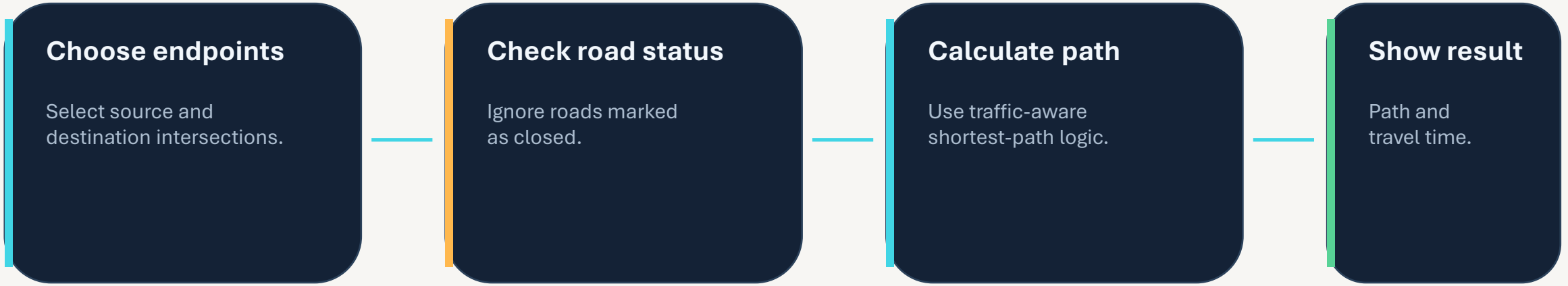
Manage closure

Close a road when unavailable or reopen it when accessible.

Updated road status is used by the route-finding process.

3. Route Calculation

Route calculation considers distance, traffic level and road availability.



If no available path exists, the system reports that no route can be found.

4. Emergency Vehicle Management

Emergency requests are stored and processed according to priority.

Register vehicle

Enter vehicle type, current intersection and destination.

Add to priority queue

Assign priority and place the vehicle in the queue.

Process next vehicle

Select the highest-priority vehicle for processing.

Priority order: Ambulance → Fire Engine → Police Vehicle

Live Demonstration Sequence

Suggested order for executing the project during the presentation.

1

Start server

Run the C++ server and open in console.

2

Show road network

Display intersections and current road conditions.

3

Update signals

Enter vehicle counts and calculate green timings.

4

Change a road

Update traffic level or close/reopen a road.

5

Find a route

Choose source and destination and show the result.

6

Process emergency

Add emergency vehicles and process the priority queue.